

Use-Case Flow Specification — Reserve Equipment Slot

Use Case ID	UC-02	Use Case Name	Reserve Equipment Slot
Primary Actor	Student	Supporting Actor	Lab Technician
Goal	Allow an authenticated student to reserve available and calibrated lab equipment for a permitted time slot.		

1. Preconditions

1. The student is authenticated in the system.
2. The required equipment is registered in the system.
3. The equipment has a valid calibration status.
4. The requested reservation is within the allowed 7-day advance period.
5. The requested reservation duration does not exceed 2 hours.

2. Main Success Scenario

1. The student selects **Reserve Equipment Slot**.
2. The system displays available lab equipment and reservation slots.
3. The student selects the required equipment and preferred time slot.
4. The system verifies the calibration status of the selected equipment.
5. The system checks whether the selected equipment and time slot are available.
6. The system validates that the reservation duration does not exceed 2 hours and is within 7 days from the current date.
7. The system creates and locks the reservation for the selected equipment and time slot.
8. The system generates a reservation confirmation/token.
9. The system displays the successful reservation confirmation to the student.
10. The use case ends successfully.

3. Postconditions

1. The selected equipment is reserved for the student.
2. The reserved time slot is no longer available for another overlapping reservation.
3. The reservation details are stored in the system.
4. A reservation confirmation/token is generated.

4. Alternate Flows

A1. Equipment or Slot Is Unavailable

- A1.1 The system detects that the selected equipment or time slot is already reserved.
- A1.2 The system displays an **“Equipment/Slot Unavailable”** message.
- A1.3 The system displays other available slots for the selected equipment.
- A1.4 The student selects an alternative available slot.
- A1.5 The system continues the reservation process from Step 6 of the Main Success Scenario.

A2. Equipment Calibration Is Invalid

- A2.1 The system detects that the selected equipment does not have a valid calibration status.
- A2.2 The system prevents the reservation from being confirmed.
- A2.3 The system informs the student that the equipment cannot currently be reserved.
- A2.4 The student selects another available calibrated equipment or exits the reservation process.